University's computer science department. Just imagine a touch-screen that multiple people could use at once, and use all 10 digits of their hands at that. Han's Frustrated Total Internal Reflectance (FTIR) multi-touch system allows just this. Using it, he can draw anything he wants on a screen, manipulate anything, and do it much faster and better than even the most competent PC professional. Yes, we can't explain it... there are multiple videos on YouTube and on his NYC homepage as well. You really have to see this to believe it. So here are the links:

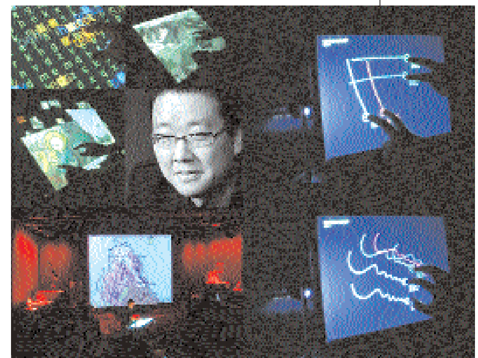
http://cs.nyu.edu/~jhan/ftirsense/
http://www.ted.com/index.php/talks/view/id/65

Apart from input devices, there's a lot being done in terms of displays as well. A lot of it we have spoken about in previous issues of *Digit*, including holographic displays, and paper or foldable / rollable displays. Looking into the distant future, technology seems to be headed the way of 3D displays, but it's still too early to speculate, and our Sixth Sense doesn't really tingle much here. Apart from holographic displays, most other 3D technologies involve wearing some sort of headset, which, honestly, is something none of us want to do. 3D Autostereoscopic Displays (no headgear needed) is still in research phase, but is eventually what we want to see. There are a few products available as well:

www.dti3d.com—Dimension Technologies develops autostereoscopic displays that can display 2D/3D visuals.

www.holografika.com—Holografika also makes autostereoscopic displays, and you can actually walk around the screen seeing shadows and aspects change to give you the true feeling of viewing a 3D object.

In a sense, the computing experience *is* all about the interface. How long can it possibly be before our keyboard/mouse combo will be called "Jurassic"? ⊠



Jeff Han recently showed off his multi-touch display of the future

Better Batteries

Alternative ways to power devices

Every family has its black sheep, every armour a chink. Batteries and hard drives have come to be the bane of technology's existence—hard drives are bottlenecks and batteries just don't last as long as they should, especially now that they are gaining in importance with everything shifting towards mobile computing, and sadly, they have been found wanting—desperately.

Enter the sugar battery. It's still a fuel cell, but in this case the fuel is anything that contains sugar—even soft drinks. If this catches on, you will be sharing your Coke with your phone and laptop. Not only do these batteries last up to four times longer than Lithium-based batteries, they're easy to recharge, are bio-degradable, and run on sugar, so you keep wondering when someone will yell "April Fool"! Such innovations are desperately required in the battery segment; the change this promises is radical, because we might soon see ourselves refilling our own batteries and never running out of charge. The developers of this technology are based at Saint Louis University, Missouri, USA, and you can read their press release at **www.slu.edu/x14605.xml**.

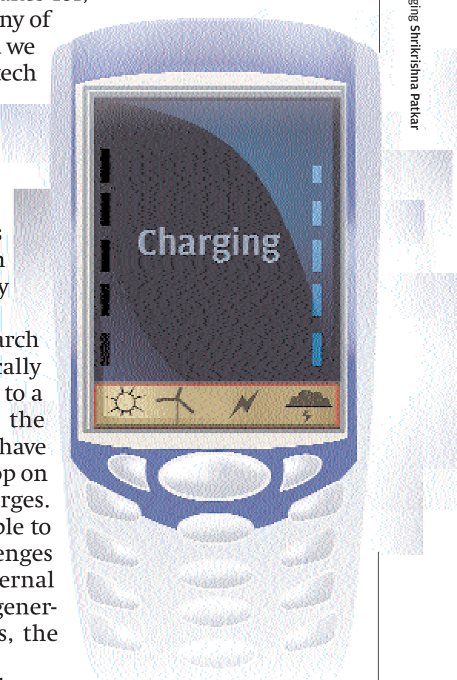
Of course, you have to understand that fuel cells are plagued with problems of their own, which is why you haven't heard much about them in recent times. For starters, they have a warm-up time, which means they cannot deliver power right away in cold conditions, which is bad for mobile devices. They also work best only at a particular rate of current pull—and this limits their use. Another major problem is longevity, which is far less than your standard Lithium-ion battery.

Researchers at MIT are investigating the use of *capacitors* as batteries! Many of you know how capacitors work—they hold charge. They can be charged in seconds. The problem with using them as batteries has been that they can't hold quite enough charge. The answer? Cover the electrodes with carbon nanotubes, which increases their surface area, and thus their charge-holding capacity. We spoke about this in our March issue, and the researchers say the technology could become commercial in five years' time. Another nanotechnological advance is that of using nanotubes in conjunction with solar cells; this makes for, well, nano solar cells. Unfortunately, many of these "five year" predictions go awry, and we can't speak with confidence about nanotech coming to the rescue.

So is there anything other than fuel cells that promise more? A magical technology like in every other segment? Sadly, no. As the chinks get ironed out, we will see fuel cells being adopted as primary batteries, but we'll still be in search of the proverbial king of battery technologies.

An absolutely different field of research is charging by induction, which basically means that you never hook up a battery to a power source; you merely place it in the vicinity of an induction charger. All you have to do is place your mobile phone or laptop on top of this pad and the battery charges. Research is underway to see if it's possible to do this at longer distances, but challenges include not damaging the delicate internal components of devices from the heat generated because of induction, and besides, the charging happens too slowly.

All this is still just stop-gap methods to our battery woes. This is the only segment of technology in dire need of radical changes. Our Sixth Sense, sadly, tells us that battery woes are only set to multiply over the coming years... ⊠



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